3.091 Exam 2 · side 1

Covalent bonding → VSEPR → orbitals → forces → bands & semiconductors (L10–L15)

CONSTANTS

hc = **1240 eV·nm** 1 eV = 1.602×10^{-19} J $N_A = 6.022\times10^{23}$

R = 8.314 J/(mol·K) 1 D = 0.2082 e·Å c = 2.998×10^{8} m/s

Visible: 1.65–3.1 eV (400–750 nm). E(eV) = 1240/λ(nm)

BOND TYPE & STRENGTH

% ionic = 100·[1 − exp(−Δχ²/4)]
μ = δ·r

Δχ <0.4 nonpolar · 0.4–1.7 polar covalent · >1.7 ionic

Stronger bond: **smaller atoms**, **higher bond order** (C≡C > C=C > C–C; HF > HCl > HBr > HI)

Ionic solid: brittle (shift → like charges repel), conducts only molten/dissolved. Covalent network: hard, high mp. Molecular: low mp (IMF).

LEWIS RECIPE

① valence e⁻ (+1 per − charge) ② centre = least EN, H never ③ singles, octets outside, leftovers on centre ④ FC = V − L − B/2 ⑤ expanded octet only period ≥ 3 (C,N,O,F never); BF_3, AlH_3, BeF_2 electron-poor ⑥ resonance = average

VSEPR: GROUPS = ATOMS + LONE PAIRS (=BOND COUNTS 1)

grp	0 LP	1 LP	2 LP	3 LP
2	linear 180°			
3	trig-plan 120°	bent <120°		
4	tetrah 109.5°	trig-pyr <109.5°	bent 104.5°	
5	TBP 90/120°	seesaw	T-shape	linear
6	octah 90°	sq-pyr	sq-planar	

LP–LP > LP–BP > BP–BP. 5 groups: LP **equatorial**; 6 groups: 2nd LP **opposite**. Name the shape by **atoms only**.

Nonpolar: identical atoms, symmetric (CO_2 BF_3 CCl_4 PCl_5 SF_6 XeF_4 AlH_3 XeF_2). **Polar**: LP breaking symmetry or unlike groups (NH_3 H_2O SF_4 IF_5 IF_3 SeO_2 CF_2H_2 HCN).

HYBRIDIZATION & Σ/Π

groups on centre: **2 sp · 3 sp² · 4 sp³ · 5 sp³d · 6 sp³d²** (lone pairs count; double/triple = 1 group)

σ: 1 per bond (head-on); π: bond order − 1 (side-on p). Triple = 1σ + 2π. #hybrids = #AOs mixed. Label gives parent shape, not exact angle (NH_3 107°, H_2O 104.5°).

C_2H_4: C sp², HOMO = π, LUMO = π*. C_2H_2: C sp.

MOLECULAR ORBITALS

n AOs → n MOs (bonding + antibonding pair). Fill lowest first, Hund.

BO = ½(e⁻ bonding − e⁻ antibonding)

Order O_2, F_2: σ2s σ*2s **σ2p π2p π*2p σ*2p**. **N_2 and lighter (B_2 C_2 N_2): π2p below σ2p.** Core 1s cancel.

species	e⁻	BO	note
H_2	2	1	
He_2 / He_2^+	4 / 3	0 / 0.5	
N_2	10	3	diamagn.
O_2	12	2	2 unpaired π*
O_2^+	11	2.5	
F_2 / Ne_2	14 / 16	1 / 0	
NO	11	2.5	1 e⁻ in π*
CO	10	3	like N_2; O lower
HCl	8	1	H1s+Cl3p; 3 LP nonbonding

More electronegative atom has **lower** AO energy (draw it lower). Nonbonding e⁻ don't count in BO. Unpaired e⁻ = paramagnetic.

BANDS & CONDUCTIVITY

n AOs → n levels → band. **Metal**: partly filled or overlapping (Na ½ full; Mg, Be: s–p overlap). **Semiconductor**: full VB, small gap ~1 eV. **Insulator**: gap > ~3 eV.

E_g(eV): C 5.4 · Si 1.1 · Ge 0.67 · Sn 0.08 · Pb 0 **(decreases down a group)**. Longer/weaker bonds → **smaller gap** (tensile strain).

σ = 1/ρ (S/m; Ω·m). Cu, Ag, Au (s^1d^{10}) best; alloys worse. Metals: σ ↓ with T. Semiconductors: σ ↑ with T (e⁻ **and** holes carry).

Photon crosses gap if hν ≥ E_g; LED λ = 1240/E_g (Si 1127 nm IR; GaN 365 nm UV). Work function KE = hν − φ.

DOPING

n-type: dopant has **more** valence e⁻ than the atom it replaces (group 15 in Si; Te) → donor level just **below** CB. **p-type**: **fewer** (group 13: B Al Ga; Zn; Ca on Ga) → acceptor level just **above** VB (holes).

Pure Si ≈ 1 carrier/10^{12} atoms; doped 1 per 10^2–10^7. Dopant fixed, carrier mobile. Too much dopant: saturation + scattering.

Dopant mass: atoms = (ρ/M)·N_A·V → ÷ (atoms per carrier, e.g. 10^5) → ÷N_A × M_{dopant}.

Compound: III–V, II–VI average 4 e⁻. E_g: GaN 3.4 · GaP 2.3 · GaAs 1.4 · GaSb 0.7 · AlN 6.3 · InSb 0.2 · ZnO 4.4 · ZnSe 2.7 · CdSe 1.7 · SiC 3.0; mixing tunes E_g.

INTERMOLECULAR FORCES

Strength: ion–ion > ion–dipole > **H-bond** > dipole–dipole > dispersion. Decision: ions? → H on N/O/F + LP? → polar shape? → else dispersion. Polar + nonpolar = dipole–induced dipole.

Dispersion ↑ with size/polarizability (CH_4 gas → $C_{17}H_{36}$ solid). BP rises with IMF: Ne < O_2 < $N(CH_3)_3$ < NaCl < CaO (2+/2− beats 1+/1−). HF high BP = H-bonds. IMF are **between** molecules, never covalent bonds.

V(r): ion–ion ∝ 1/r, ion–dipole 1/r^2, dipole–dipole 1/r^3.

3.091 Exam 2 · side 2

Crystals → diffraction → defects → glass & polymers → diffusion (L16–L20)

CRYSTAL = LATTICE + MOTIF

Lattice: periodic array of **equivalent** points; motif/basis: what sits at each. Honeycomb = hex lattice + 2-atom motif. 7 systems / **14 Bravais** (3 cubic, 2 tetr., 4 orthorh., 2 monocl., 1 hex., 1 rhombo., 1 tricl.). 2D: oblique, rect., square, hex.

Cubic a=b=c, 90°. Tetr. a=b≠c. Hex γ=120°. Rhombo a=b=c, α=β=γ≠90°.

CUBIC CELLS

	SC	BCC	FCC
pts/cell	1	2	4
atoms touch along	edge	body diag.	face diag.
a(r)	2r	4r/√3	2√2 r
packing	0.52	0.68	0.74
nn dist.	a	a√3/2	a/√2
# nn	6	8	12

$\rho = (\text{points}\cdot M) / (N_A \cdot a^3)$
packing = $N\cdot(4/3)\pi r^3 / a^3$

Corner 1/8, face 1/2, edge 1/4, body 1. FCC = ABCABC (CCP), HCP = ABAB; both 74%, 12 nn.

STRUCTURES

NaCl: FCC, 4Na+4Cl (8 ions), oct. holes. **CsCl**: **simple cubic**, 2-ion motif, 8-fold. **ZnS**: FCC S, Zn in ½ tetr. holes (8 ions). **CaF_2**: FCC, 4Ca+8F (12). **$CaTiO_3$** perovskite: Ca corners, Ti centre, O faces. Per FCC cell: 4 oct., 8 tetr. holes. 1:1 ratio necessary, not sufficient for rock salt.

Au: a = 407 pm → r = 144 pm. Fe BCC a = 2.86 Å (r 1.24). Cu FCC a = 3.61 Å.

MILLER INDICES & FAMILIES

Intercepts (in a,b,c) → reciprocals → clear fractions → (hkl). ∞→0. Negative: bar. (hkl) plane, {hkl} family, [uvw] direction, <uvw> family, point x,y,z. Cubic only: (hkl) ⊥ [hkl].

family	count
<100>	6 dirs
<110>	12 dirs
<111>	8 dirs
{100}	6 faces (3 orient.)
{110}	6 planes
{111}	4 planes

Planar density: sketch plane, atoms inside (corner ¼, edge ½), ÷ area. BCC (110): 2 atoms / ($a\cdot a\sqrt{2}$). FCC (100): 2 / a^2. FCC (111) densest, BCC (110) densest. Slip: FCC {111}<110>; BCC {110} <111>.

DIFFRACTION

$d_{hkl} = a/\sqrt{(h^2+k^2+l^2)}$ (cubic)
$2d \sin\theta = n\lambda$

θ = glancing angle; patterns plot **2θ** → halve! Cu Kα 1.54 Å = 0.154 nm. Calc in degree mode. $d_{220} = d_{110}/2$. r (atom), a (cell edge), d (plane spacing) are different.

Selection rules: SC all · **BCC h+k+l even** (110, 200, 211, 220…) · **FCC all odd or all even** (111, 200, 220, 311, 222…).

$\sin^2\theta \propto h^2+k^2+l^2$: SC 1:2:3:4:5:6:8 · BCC 2:4:6:8:10:12 · FCC **3:4:8:11:12:16**. Powder peaks give structure; peak angle gives a.

Example: λ=1.54 Å, Fe BCC a=2.866 → d_{110}=2.03, θ=22.3°, 2θ=44.7°.

DEFECTS

Point (0D): vacancy · interstitial (small atom, C in Fe) · substitutional (similar size, Zn in Cu, dopants). Ionic: **Schottky** = cation + anion vacancy pair; **Frenkel** = ion moves to a gap (vacancy + interstitial, usually cation). **Line (1D)**: edge dislocation ⊥ = extra half-plane; moves along slip plane → plastic deformation at 10^3–10^4× lower stress. **Planar (2D)**: grain boundary; polycrystal vs single crystal.

Work hardening: more, tangled dislocations → harder, more brittle. **Anything blocking dislocations hardens** (alloy atoms, C in steel, hammering). **Annealing**: heat (no melting) → diffusion removes dislocations → softer, tougher.

GLASS & POLYMERS

Amorphous: no long-range order; forms if structure packs badly or cools fast. SiO_4 tetrahedron: Si–O 162 pm, 109.5°; quartz periodic, glass random. **Network modifiers** (Na, Ca) → non-bridging O → lower softening T (soda-lime < borosilicate < 96% silica < fused silica). No sharp melting: viscosity falls smoothly.

Faster cooling → **higher molar volume, higher T_g**; crystal drops suddenly at T_m. Polymers: monomers → chains; crosslink (S, vulcanization) → does not melt; HDPE semicrystalline.

DIFFUSION

$J = -D\, dC/dx$ $D = D_0 \exp(-Q/RT)$
$x \approx \sqrt{(Dt)}$

ln D vs 1/T: slope = **−Q/R** (steeper = larger Q; use J/mol). $Q = R \ln(D_2/D_1)/(1/T_1-1/T_2)$. D ↑ strongly with T. **Interstitial** (C, H in Fe): low Q, fast. **Vacancy / self** (Fe in Fe): high Q, Q_v ↑ with T_m. Fick 2nd: $\partial C/\partial t = D\, \partial^2 C/\partial x^2$ (profile changes with time); surface at fixed C → erfc profile; depth ∝ √t.

Flux ×area = rate (doors: $D\cdot\Delta C/\Delta x\cdot A$).